

Summary and Recommendations

Objective

The objective of this analysis is to:

- Identify key factors influencing customer churn
- Understand how different services and demographics impact retention
- Use visualizations to uncover patterns and trends
- Provide actionable insights to reduce churn and improve customer loyalty

Visualizations & Approach

To explore the dataset, multiple **countplots (with churn segmentation)** were created across key categorical features:

- Demographics: **SeniorCitizen**
- Services:
 - PhoneService, MultipleLines
 - InternetService
 - OnlineSecurity, OnlineBackup, DeviceProtection
 - TechSupport
 - StreamingTV, StreamingMovies

Visualization Strategy:

- Used **subplots (grid layout)** to compare multiple features simultaneously
- Applied **hue = Churn** to clearly distinguish churn vs non-churn customers
- Rotated labels and optimized layout for readability
- Focused on **pattern recognition across categories**

These visualizations made it easy to identify **high-risk customer segments** and compare service-level impact on churn.

Key Insights & Findings

1. Senior Citizens (High-Risk Segment)

- Senior citizens contribute disproportionately to churn
- **Churn rate: ~40–45% vs ~25–30% (non-seniors)**
- ~1.5x higher likelihood of churn

2. Internet Service (Strongest Impact)

- **Fiber Optic users: ~40–45% churn (highest)**
- DSL users: ~20–25% churn
- No internet: <10% churn

Fiber users churn at nearly **double the rate of DSL users**

3. Phone Service & Multiple Lines

- Phone service alone has minimal impact
- **Multiple lines users: ~30–35% churn**
- Single-line users: ~25–28% churn

Slight increase suggests **bundle dissatisfaction or pricing concerns**

4. Add-On Services (Retention Drivers)

Customers without add-ons show significantly higher churn:

- **Online Security**
 - No: ~40–45% churn
 - Yes: ~15–20%
- **Online Backup**
 - No: ~35–40%
 - Yes: ~20–25%
- **Device Protection**
 - No: ~35–40%
 - Yes: ~20–25%

Add-ons reduce churn by **15–25 percentage points**

5. Tech Support (Critical Factor)

- No Tech Support → **~40–45% churn**
- With Tech Support → **~15–20% churn**

One of the **strongest churn reducers (~2x improvement)**

6. Streaming Services (Moderate Impact)

- Streaming TV & Movies show **moderate churn variation (~25–35%)**
- Not primary drivers, but contribute in combination with other factors

Data-Driven Insights

- Customers with **fewer services** → churn ~35–45%
- Customers with **multiple services** → churn drops to ~15–25%
- High churn clusters around:
 - Fiber optic users
 - Customers without support services
 - Low engagement users

Clear pattern: **Higher engagement = Lower churn**

Recommendations

1. Improve Fiber Optic Service

- Address high churn (~45%) by:
 - Reviewing pricing structure
 - Improving service reliability
 - Offering retention discounts

2. Bundle Add-On Services

- Combine:
 - Security + Backup + Protection
- Offer discounted packages or trials
- Can reduce churn by up to **25%**

3. Strengthen Tech Support

- Expand access to:
 - 24/7 support
 - Priority issue resolution

- Potential to cut churn nearly in half

4. Target High-Risk Segments

- Focus on:
 - Senior citizens
 - Fiber users
- Provide:
 - Personalized plans
 - Better onboarding & assistance

5. Increase Customer Engagement

- Encourage adoption of multiple services
- Use:
 - Cross-selling strategies
 - Personalized recommendations

Conclusion

The analysis highlights that churn is **not random**, but strongly influenced by:

- Service quality
- Customer support
- Level of engagement
- **High churn (~40–45%)** → low engagement, no support, fiber users
- **Low churn (~15–25%)** → bundled services, strong support

By optimizing service offerings and improving customer experience, churn can be **reduced by 20–25 percentage points**.